

Notes: Zhong and Zhou (RFS, 2026)

Dynamic Coordination and Bankruptcy Regulations

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1 Big picture

- **Question.** How do bankruptcy regulations that restrict creditors' ability to collect *ex post* affect creditors' incentives to stay invested *ex ante*? In particular, when do early bankruptcy filing, automatic stay, and avoidable preference improve coordination, and when can they backfire?
- **Setting.** The paper studies a distressed but still productive firm with staggered debt, asynchronous creditor learning, and endogenous exit timing. Creditors privately learn that a bad shock has reduced the firm's growth rate, and then decide when to stop rolling over their debt.
- **Core challenge.** Bankruptcy regulation is often designed to improve *ex post* coordination by stopping a first-come-first-serve scramble for assets. But this same protection can weaken creditors' *ex ante* incentives to stay in the firm, because more creditors know they may be trapped in bankruptcy.
- **Main conceptual contribution.** The paper isolates two forces that determine creditors' waiting behavior:
 - the **recovery rate channel**: how much a creditor expects to recover if bankruptcy arrives before exit;
 - the **hazard rate channel**: how likely it is that the creditor will be caught by bankruptcy before exiting.

Almost every result in the paper is a trade-off between these two channels.

- **Main theoretical insight 1.** A firm can sometimes prolong its life by committing to *earlier* bankruptcy filing. Filing earlier preserves assets for latecomers, which raises recovery in bankruptcy and can make creditors more willing to stay invested before bankruptcy actually occurs.
- **Main theoretical insight 2.** Regulators' clawback of prebankruptcy repayments (*avoidable preference*) can dominate the firm's own commitment to file early. The reason is that clawback can shift forward the pivotal creditor who is allowed to exit with full repayment *without* terminating production at that earlier point.
- **Main regulatory insight.** The welfare-maximizing regulator wants earlier intervention than the firm does. The regulator cares not only about delaying bankruptcy, but also about keeping productive capital inside the firm rather than allowing withdrawals that dissipate going-concern value.
- **Institutional applications.** The model gives a unified way to think about:
 - **automatic stay** in corporate bankruptcy,
 - **avoidable preference** / repayment clawbacks,
 - **bank failures** and the timing of regulatory seizure,
 - **seniority structure** among creditors.
- **Key policy result.** The optimal clawback window is “detail-free” in the sense that it does not depend on the firm's production parameters such as initial asset level or post-shock growth rate; instead, it depends on the interest rate, shock intensity, and debt-maturity / information-transmission structure.

- **Main extension result.** Ex ante seniority does not necessarily improve coordination. A small share of senior creditors can worsen runs by making junior creditors less patient; only when the senior class becomes sufficiently large can its greater patience delay bankruptcy.
- **Economic takeaway.** Bankruptcy regulation is not simply about stopping runs after they begin. It also changes the incentives that determine *when* runs start. The best policy balances better recovery against a higher hazard of being drawn into bankruptcy.

2 Data and measurement (key objects)

2.1 This is a theory paper, so the “data” are institutional facts and model objects

- The paper does *not* estimate the model structurally on a micro dataset. Instead, it develops a dynamic theory discipline around bankruptcy institutions and then interprets the model in the context of actual legal provisions and bank-failure episodes.
- The core empirical motivation is institutional:
 - **Automatic stay** suspends individual debt collection after a bankruptcy filing.
 - **Avoidable preference** claws back repayments made shortly before bankruptcy and redistributes them in court.
 - Bank regulators often seize failing banks while considerable assets are still present, rather than waiting until assets are fully depleted.
- The paper notes that between 2017 and 2019, avoidable preference was cited in roughly half of the U.S. bankruptcy cases in a Westlaw sample, and it also discusses typical clawback windows across the United States, China, and the United Kingdom. This motivates taking repayment clawbacks seriously as a first-order institution rather than as a legal footnote.

2.2 Main primitives in the model

- A : initial asset value of the firm.
- Debt face value is normalized to 1 at date 0.
- g : preshock asset-growth rate and promised interest rate on debt.
- $g' \in (0, g)$: post-shock growth rate after distress arrives.
- λ : Poisson intensity of the bad shock.
- η : debt maturity / information-transmission window; creditor i learns of the shock over the interval $[t_0, t_0 + \eta]$.
- $k \in (0, 1)$: bankruptcy threshold, interpreted as the fraction of creditors allowed to exit before bankruptcy.
- $m \geq 0$: clawback window under avoidable preference.
- τ : waiting time after learning the bad shock.

Interpretation. These are the key comparative-statics objects. Everything the paper studies comes from how changes in k and m alter the equilibrium waiting time τ^* through recovery and hazard.

2.3 Key endogenous objects

- **Exit timing.** In the symmetric linear equilibrium, each creditor exits at

$$\beta_i(t_i) = t_i + \tau.$$

- **Exit rate.** Once the bad shock is known by a positive mass of creditors, exits occur gradually at rate $1/\eta$ over the relevant window.
- **Bankruptcy time.** Bankruptcy is triggered when a fraction k of creditors has exited.
- **Recovery payoff.** Creditors who do not exit in time receive a pro-rata bankruptcy payoff, which depends on how many assets remain inside the firm and, under avoidable preference, on how many repayments are clawed back.
- **Hazard rate.** The hazard rate measures the chance that a creditor is caught by bankruptcy before exiting successfully.
- **Welfare.** Welfare is total creditor payoff. Because the distressed firm eventually goes bankrupt in equilibrium, equity is not the relevant claimant in the benchmark welfare analysis.

2.4 Information structure and why it matters

- Creditors learn about the bad shock *asynchronously*. This is the clock-game feature inherited from Abreu–Brunnermeier type models.
- Actions are private. When deciding how long to wait, a creditor does *not* observe how many others already know the bad news or have already withdrawn.
- This is why timing matters so much. The run is not a one-shot decision of “run or stay”; it is a dynamic problem of *when* to exit.
- The asynchronous structure is natural for both corporate debt and bank runs:
 - in corporations, creditors may learn when debt matures or covenants are tested;
 - in banking, uninsured depositors may learn gradually about losses or duration risk.

2.5 Why these objects are enough for the paper’s purposes

- The paper is trying to understand policy design, not estimate a reduced-form treatment effect.
- For that objective, the central quantities are the *pivotal creditor*, the *timing of bankruptcy*, the *recovery payoff*, and the *hazard of being caught in bankruptcy*. Once those are pinned down, the model can compare alternative legal rules cleanly.
- The paper also deliberately abstracts from public signals, noisy asset prices, and renegotiation with all creditors. These omissions make the model more transparent and keep attention on the recovery-versus-hazard trade-off.

3 Models and empirical specifications (all equations + interpretation)

3.1 Benchmark model: environment, timing, and shock structure

Time is continuous. The firm starts with assets A and liabilities normalized to one. Assets and liabilities both grow at rate g before distress arrives, so the leverage ratio is constant. At a random time $t_0 > 0$, a bad shock arrives with Poisson intensity λ and reduces the firm's asset growth rate from g to $g' \in (0, g)$.

Creditors have staggered maturity. Creditor $i \in [0, 1]$ learns the shock at

$$t_i = t_0 + i\eta.$$

Conditional on learning time t_i , the posterior density over the shock date is

$$\psi(t_0 | t_i) = \frac{\lambda e^{-\lambda(t_0 - t_i)}}{e^{\lambda\eta} - 1}, \quad t_0 \in [t_i - \eta, t_i]. \quad (1)$$

Interpretation. Even after learning the firm is in trouble, a creditor still does not know exactly when trouble started. That uncertainty is what generates the dynamic coordination problem.

3.2 Exit rate, asset dynamics, and bankruptcy time

Let w_t denote the exit rate of creditors at time t . The firm's assets evolve as

$$dY_t = \begin{cases} gY_t dt, & t \leq t_0, \\ (g'Y_t - w_t e^{gt}) dt, & t > t_0. \end{cases} \quad (2)$$

If a fraction k of creditors has exited, bankruptcy occurs. The bankruptcy time is

$$\hat{t}(k, t_0, \{\beta_i\}) = \inf \left\{ u : \int_{t_0}^u w_t(t_0, \{\beta_i\}) dt \geq k \right\}. \quad (3)$$

Interpretation.

- Before the shock, the firm simply grows.
- After the shock, the firm still grows, but more slowly, and it also bleeds cash because exiting creditors are repaid.
- Bankruptcy is therefore not triggered mechanically by calendar time; it is triggered by the *mass of withdrawals*.

3.3 Creditor payoff, manager objective, and welfare

A creditor who exits before bankruptcy receives full repayment $e^{g\beta_i}$. A creditor caught by bankruptcy receives a recovery payoff

$$\frac{Y_{\hat{t}}}{1 - k}.$$

Given threshold k and the strategy of other creditors, creditor i chooses β_i to maximize

$$\begin{aligned} \Pi_i(\beta_i \mid t_i, \beta_{-i}^*, k) = & \int_{\{\beta_i \leq \hat{t}(t_0)\}} e^{g\beta_i} \psi(t_0 \mid t_i) dt_0 \\ & + \int_{\{\beta_i > \hat{t}(t_0)\}} \frac{Y_{\hat{t}}}{1-k} \psi(t_0 \mid t_i) dt_0. \end{aligned} \quad (4)$$

The manager chooses the bankruptcy threshold k to maximize firm lifespan. With commitment,

$$k^c = \arg \max_k \hat{t}(k, t_0, \{\beta_i^* \mid k\}), \quad (5)$$

while without commitment bankruptcy is triggered only when remaining assets are exhausted.

Welfare is defined as total creditor payoff:

$$W = \int_{t_0}^{t_0 + \eta} \Pi_i(\beta_i^* \mid t_i, \beta_{-i}^*, k) f(t_i \mid t_0) dt_i. \quad (6)$$

Interpretation.

- The manager wants to maximize survival time.
- The regulator wants to maximize total creditor payoff, which includes preserving productive capital and controlling capital outflow.
- This wedge between manager and regulator objectives is central to the policy results.

3.4 Symmetric linear equilibrium and Assumption 1

The paper focuses on symmetric linear equilibria:

$$\beta_i(t_i) = t_i + \tau. \quad (7)$$

That is, every creditor waits a common duration τ after learning the bad shock.

To ensure existence and uniqueness of the relevant equilibrium, the paper imposes **Assumption 1**, which requires:

- assets cannot be too large relative to liabilities;
- maturity / the learning window η must be sufficiently long, specifically $\eta > 1/(g - \lambda) > 0$.

Interpretation.

- If assets are too abundant, the coordination problem becomes weak.
- If debt matures too quickly or information spreads too fast, early exits no longer impose large externalities on late creditors.

3.5 Recovery payoff and decomposition of waiting time

In a symmetric equilibrium, the pivotal creditor exits at

$$\hat{t}(k, t_0, \beta) = t_0 + k\eta + \tau. \quad (8)$$

The remaining assets at bankruptcy imply the recovery payoff

$$\frac{Y_t}{1-k} = \frac{Ae^{g'(\tau+k\eta)} - \frac{e^{g(\tau+k\eta)} - e^{g\tau+g'k\eta}}{(g-g')\eta}}{1-k} e^{gt_0} \equiv \alpha(\tau, k) e^{gt_0}. \quad (9)$$

The creditor's first-order condition yields a key decomposition. Define the hazard rate

$$h(k) = \frac{\psi(t_0 = t_i - k\eta \mid t_i)}{1 - \Phi(t_0 = t_i - k\eta \mid t_i)} = \frac{\lambda e^{\lambda k\eta}}{e^{\lambda k\eta} - 1}. \quad (10)$$

Then **Proposition 1** states that any interior waiting time satisfies

$$\tau^* = \frac{1}{g} \log(\alpha(\tau^*, k) e^{-gk\eta}) + \frac{1}{g} \log\left(\frac{h(k)}{h(k) - g}\right). \quad (11)$$

Interpretation.

- The first term is the **recovery rate channel**: higher recovery in bankruptcy makes creditors more willing to wait.
- The second term is the **hazard rate channel**: a higher chance of being caught by bankruptcy makes creditors less willing to wait.
- This decomposition is the conceptual heart of the paper.

3.6 Closed-form equilibrium waiting time

Proposition 2 shows that the equilibrium waiting time can be written as

$$\tau^*(k) = \max \left\{ 0, \frac{1}{g-g'} (\log A - \log v(k)) \right\}, \quad (12)$$

where

$$v(k) = \frac{e^{(g-g')k\eta} - 1}{(g-g')\eta} + (1-k) \left[\frac{g}{\lambda} e^{(g-\lambda-g')k\eta} - \left(\frac{g}{\lambda} - 1 \right) e^{(g-g')k\eta} \right]. \quad (13)$$

Interpretation. The corner $\tau^* = 0$ is possible. If hazard is sufficiently high or recovery sufficiently low, everyone exits immediately upon learning the bad shock.

3.7 Bankruptcy threshold without commitment

Without commitment, bankruptcy occurs only when assets are exhausted:

$$Y_{t_0+k^{nc}\eta+\tau^*(k^{nc})} = 0.$$

Proposition 3 then shows

$$k^{nc} = \frac{1}{(g-g')\eta} \log(A(g-g')\eta + 1), \quad (14)$$

and the corresponding equilibrium waiting time is

$$\tau^*(k^{nc}) = 0.$$

Interpretation. If bankruptcy only occurs after every remaining asset is drained, late creditors get zero recovery. Anticipating this, creditors run immediately. So the apparent ex post flexibility of “wait until assets are gone” is actually destructive ex ante.

3.8 Bankruptcy threshold with commitment

With commitment, the manager chooses k to maximize

$$k^c = \arg \max_k [k\eta + \tau^*(k)]. \quad (15)$$

Proposition 4 shows that the manager sometimes chooses an *interior* threshold $k^c < k^{nc}$, meaning bankruptcy is filed *early*, before assets are fully exhausted. This happens when:

- the shock is relatively mild (higher g'),
- leverage is relatively low (higher A),
- so the recovery-rate channel is strong enough to dominate the mechanical benefit of letting more creditors exit.

Interpretation.

- A higher k delays bankruptcy mechanically.
- But it also reduces assets left in bankruptcy, worsening recovery and potentially causing creditors to run sooner.
- Therefore, survival time can be hump-shaped in k .

3.9 Application: automatic stay

The commitment case corresponds to a world with **automatic stay**: once bankruptcy is filed, creditors can no longer race individually to seize assets. The no-commitment case corresponds to an informal first-come-first-serve world in which latecomers are left with nothing.

Interpretation.

- Automatic stay can increase ex ante patience because it preserves some bankruptcy value.
- The model predicts it is especially valuable when the recovery-rate channel is strong, that is, when shocks are milder and leverage is lower.

3.10 Welfare-maximizing bankruptcy threshold and bank-failure application

In the linear equilibrium, welfare can be written as

$$W(k) = \int_{t_0+\tau^*}^{t_0+k\eta+\tau^*} \frac{1}{\eta} e^{gt} dt + Y_{t_0+k\eta+\tau^*}. \quad (16)$$

Proposition 5 states:

$$k^W \leq k^c, \quad (17)$$

with strict inequality in the parameter region where early bankruptcy is valuable.

Interpretation.

- The manager only wants to maximize life.

- The regulator wants to maximize life *and* reduce the amount of productive capital that leaves the firm before bankruptcy.
- Hence the regulator prefers even earlier intervention than the manager.

This also gives a bank-run implication: a regulator may optimally seize a bank before the bank would voluntarily fail, because preserving more assets for uninsured depositors can reduce run incentives.

3.11 Avoidable preference: clawback window and modified recovery

The regulator can choose a clawback window m . Creditors who exit at least m periods before bankruptcy keep full repayment. Creditors paid inside the last m periods before bankruptcy must return those repayments and instead share in the bankruptcy pool.

The bankruptcy payoff becomes

$$\alpha(\tau, k, m)e^{gt_0} = \frac{Ae^{g'(k\eta+\tau)} - \frac{e^{g\tau}}{(g-g')\eta} (e^{gk\eta} - e^{g'k\eta}) + \frac{e^{g\tau}}{g\eta} (e^{gk\eta} - e^{g(k\eta-m)})}{1 - k + \frac{m}{\eta}} e^{gt_0}. \quad (18)$$

With clawback, the creditor first-order condition implies the new hazard rate

$$h(k, m) = \frac{\lambda e^{\lambda(k\eta-m)}}{e^{\lambda(k\eta-m)} - 1}, \quad (19)$$

and the new waiting-time decomposition

$$\tau^* = \frac{1}{g} \log \left(\alpha(\tau^*, k, m) e^{-g(k\eta-m)} \right) + \frac{1}{g} \log \left(\frac{h(k, m)}{h(k, m) - g} \right). \quad (20)$$

Interpretation.

- Clawback improves recovery because more repayments are pulled back into the bankruptcy estate.
- But clawback also makes successful exit harder, because now a creditor must exit *farther* ahead of bankruptcy to truly keep the money.
- Again, the recovery-rate and hazard-rate channels move in opposite directions.

3.12 Why maximizing welfare is equivalent to maximizing waiting time

Proposition 6 shows that, for given k , total welfare is increasing in τ^* , so the regulator's problem can be written as

$$m^* = \arg \max_{m < k\eta} W(k, m) = \arg \max_{m < k\eta} \tau^*(k, m). \quad (21)$$

Interpretation. Under avoidable preference, clawback is a pure redistribution of repayment claims ex post. Its social value comes entirely from how it changes creditors' ex ante patience.

3.13 Optimal clawback window

Proposition 7 gives the optimal clawback window:

$$m^*(k) = \max \left\{ 0, \frac{1}{g - \lambda} - (1 - k)\eta \right\} < k\eta. \quad (22)$$

Equivalently,

$$k - \frac{m^*(k)}{\eta} = 1 - \frac{1}{(g - \lambda)\eta}. \quad (23)$$

Interpretation.

- The regulator effectively chooses the *pivotal creditor* who is allowed to escape clawback.
- A higher shock intensity λ or faster information transmission (smaller η) strengthens the hazard channel, which calls for a *longer* clawback window.
- A higher interest rate g raises the size of the repayment base that can be clawed back, strengthening the recovery channel and allowing a *shorter* clawback window.

3.14 Avoidable preference without managerial commitment

Proposition 8 shows that with optimal clawback, even a manager *without* commitment can induce more creditor patience than in the benchmark commitment case without clawback. The equilibrium is

$$k_{nc}^m = \frac{1}{g\eta} \log \left(\frac{g^2}{(g - \lambda)\lambda} e^{(g - \lambda)(\eta - \frac{1}{g - \lambda})} - \frac{g - \lambda}{\lambda} e^{g(\eta - \frac{1}{g - \lambda})} \right), \quad (24)$$

$$\tau^* = \frac{1}{g - g'} \left[\log A - \log \left(\frac{e^{(g - g')k_{nc}^m \eta} - 1}{(g - g')\eta} \right) \right], \quad (25)$$

$$m^* = k_{nc}^m \eta - \left(\eta - \frac{1}{g - \lambda} \right). \quad (26)$$

Moreover,

$$\tau^*(k_{nc}^m, m^*) \geq \tau^*(k^c, 0) \geq \tau^*(k^{nc}, 0) = 0, \quad (27)$$

with strict inequality when early bankruptcy is valuable in the benchmark.

Interpretation.

- Clawback can replicate the same pivotal creditor as an early-filing rule.
- But clawback lets production continue after that pivotal point, so the same hazard is combined with a *higher* recovery rate.
- That is why clawback is strictly better than commitment to early filing in the relevant region.

3.15 Redundancy of managerial commitment under avoidable preference

Proposition 9 states that under avoidable preference,

$$k_c^m = k_{nc}^m,$$

and the equilibrium m^* and τ^* are the same whether or not the manager can commit to a bankruptcy threshold.

Interpretation. Once the regulator can claw back repayments ex post, the firm’s own commitment technology becomes redundant. Ex post redistribution can implement the desired pivotal creditor without the deadweight loss of shutting down production too early.

3.16 Sequence of actions

The main analysis assumes:

1. the firm chooses k ,
2. the regulator chooses m ,
3. creditors choose τ .

The paper then shows that the core result—that ex post clawback dominates early filing—is robust to alternative timings:

- regulator first, firm later;
- simultaneous choice of k and m ;
- more complicated leader–follower timelines.

Interpretation. The exact closed form for m^* may be lost when the regulator internalizes how m changes the firm’s best response for k , but the *ranking* of avoidable preference over simple early-filing commitment survives.

3.17 Extension: seniority

Suppose a fraction ω of creditors is senior and the rest is junior. Let τ_S and τ_J denote their waiting times.

Proposition 10 shows:

- juniors always run faster than seniors: $\tau_J^* \leq \tau_S^*$;
- there is a cutoff $\tilde{\omega} \in (0, 1 - k)$:
 - if $\omega \leq \tilde{\omega}$, seniors do not exit before bankruptcy and receive full payment in bankruptcy;
 - if $\omega > \tilde{\omega}$, seniors also exit before bankruptcy and both groups can be pivotal.

Interpretation.

- A small number of senior creditors hurts coordination, because juniors know those seniors will absorb more of the recovery value.
- When the senior class becomes large enough, seniors themselves become pivotal, and their greater patience can delay bankruptcy.
- So the effect of seniority is nonmonotonic.

3.18 Extension: recoverable bad shock

The paper also allows the bad shock to reverse. At a Poisson time $\tilde{\tau}_R$, the growth rate returns from g' to g .

In that case, if recovery happens before exits occur, creditors optimally stay until natural maturity and the regulator again chooses m to maximize waiting time.

Interpretation.

- Recovery introduces a new benefit of patience: creditors may avoid a run entirely if they wait long enough for the shock to dissipate.
- This does not overturn the main clawback result. If anything, it reinforces the value of policies that preserve production without forcing immediate shutdown.

4 Identification and instruments (what makes the results credible)

4.1 What the theory pins down

- The model is built to isolate a clean dynamic trade-off:

wait longer $\iff$ gain interest income but risk bankruptcy.

- The recovery-rate channel is identified by what assets remain inside the firm at bankruptcy.
- The hazard-rate channel is identified by where the creditor lies relative to the pivotal creditor.
- Automatic stay, early bankruptcy, and clawback are all modeled as mechanisms that move those two channels in different ways.

4.2 Why the clock-game structure is useful

- Static global-game models let agents choose whether to run, but not exactly when.
- Here, timing is the entire object of interest. The paper wants to know how the timing of exit changes the evolution of the asset pool and the chance of successful withdrawal.
- That makes the dynamic clock-game environment the right laboratory for studying bankruptcy rules that affect who can leave first and what remains for latecomers.

4.3 Why the welfare criterion reduces to waiting time under clawback

- Proposition 6 is economically important because it converts the regulator's problem into a one-dimensional object: make creditors more patient.
- This is not a trivial simplification. It relies on the fact that clawback reallocates value rather than creating new value directly; it changes welfare only by changing the equilibrium timing of withdrawals.
- That result is what makes the closed-form optimal clawback window possible.

4.4 What makes the paper’s policy insights credible

- The model nests two realistic institutional margins:
 - **timing of bankruptcy filing**,
 - **treatment of prebankruptcy repayments**.
- The two central regulations—automatic stay and avoidable preference—map directly into model objects:
 - automatic stay $\leftrightarrow$ commitment to a bankruptcy threshold;
 - avoidable preference $\leftrightarrow$ repayment clawback window m .
- The comparative statics are transparent. Stronger shocks, faster information transmission, and larger interest payments have intuitive and closed-form effects on optimal clawback.

4.5 Main limitations

- There is no empirical estimation of the model, so the paper is strongest on mechanism and weaker on quantitative magnitude.
- Exits are private and there are no public price signals. This is convenient but may understate the information feedback present in public debt markets.
- The bad shock is a reduced-form drop in asset growth; the model does not explain where the shock comes from.
- The framework abstracts from explicit legal and administrative costs of enforcing clawback, though the paper is explicit that those costs matter for real-world policy ranking.
- Equity / managerial agency is simplified. The manager’s objective is simply to maximize firm life in the benchmark.

4.6 Relation to the literature

- **Clock-game literature.** The paper extends the Abreu–Brunnermeier tradition from bubbles and crashes to bankruptcy and creditor runs.
- **Global-game bank-run literature.** Relative to static run models, the paper emphasizes exact exit timing and how that timing affects both asset dynamics and recovery payoffs.
- **Bankruptcy literature.** The paper contributes by focusing on ex post legal rules—automatic stay and avoidable preference—and studying how they alter ex ante run incentives.
- **Policy literature on runs.** The paper belongs to the class of models where policy changes payoffs directly, but it adds a new policy instrument: repayment clawback.

5 Tables and figures

5.1 Figure 1: equilibrium waiting time

Read:

- Figure 1 plots the fixed-point determination of τ^* in Proposition 1.
- The recovery-rate term is downward sloping in waiting time because longer waiting lets more capital leave before bankruptcy, reducing what is left for latecomers.
- The hazard-rate term is flat in the symmetric equilibrium because the common waiting time does not change who the pivotal creditor is.
- The intersection with the 45° line gives the unique equilibrium waiting time.

Economic message. The figure makes the paper's central trade-off visual: more waiting can be self-defeating because it lowers recovery even while generating more interest income.

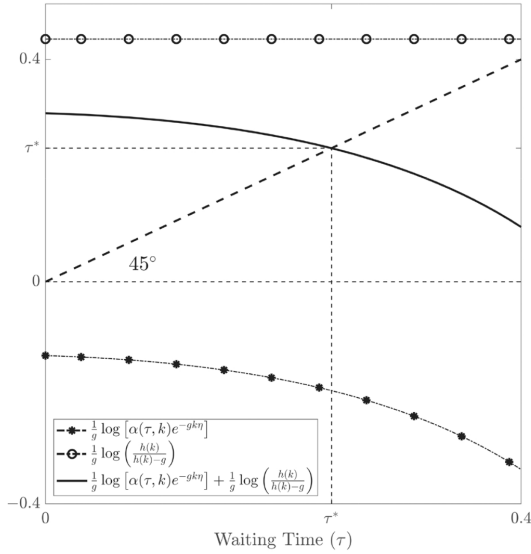


Figure 1
Equilibrium waiting time
The figure depicts the effect of waiting time τ on the recovery rate channel and the hazard rate channels, and the determination of τ^* as the fixed-point solution to (16). The decreasing curve with solid dots reflects the recovery rate channel (i.e., $\frac{1}{g} \log(\alpha(\tau, k)e^{-gk\eta})$). The flat line with circles plots the hazard rate channel (i.e., $\frac{1}{g} \log\left(\frac{h(k)}{h(k)-g}\right)$). The combined effect given by the solid curve intersects with the 45-degree line, giving the equilibrium waiting time τ^* . The parameters for this figure are $A=1.1$, $g=2$, $\lambda=0.2$, $\eta=0.6$, $g'=1$, and $k=0.5$.

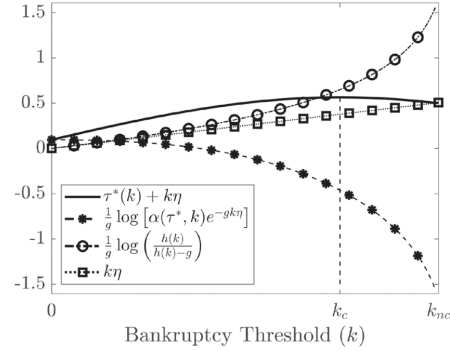


Figure 2
Bankruptcy threshold and the firm's lifespan
We decompose the equilibrium lifespan of the firm after the arrival of a bad shock (i.e., $\tau^*(k) + k\eta$, the unmarked curve) into the decreasing recovery rate channel (i.e., $\frac{1}{g} \log(\alpha e^{-gk\eta})$, the dashed curve with dots), the increasing hazard rate channel (i.e., $\frac{1}{g} \log\left(\frac{h(k)}{h(k)-g}\right)$, the dashed curve with circles) and the mechanical delay components (i.e., $k\eta$, the dotted curve with squares) according to (16). The parameters for this figure are $A=1.1$, $g=2$, $\lambda=0.2$, $\eta=0.6$, and $g'=1$.

5.2 Figure 2: bankruptcy threshold and firm lifespan

Read:

- Figure 2 decomposes total survival time $k\eta + \tau^*(k)$ into:
 - the mechanical delay from allowing more creditors to exit ($k\eta$),
 - the hazard-rate effect,
 - the recovery-rate effect.

- The overall curve is hump-shaped. That is the visual proof that early filing can prolong firm life.

Economic message. A larger bankruptcy threshold is not always better. Beyond some point, the loss in recovery dominates the extra time gained from letting more creditors exit.

5.3 Figure 3: when early filing matters more

Read:

- Figure 3 varies the post-shock growth rate g' and the asset level A .
- When the shock is milder or the firm is less leveraged, the interior optimum for k is more likely to exist.
- When the shock is severe or leverage is high, the manager just lets creditors deplete assets and the no-commitment region dominates.

Economic message. Automatic stay and early filing are most useful when there is enough going-concern value worth preserving.

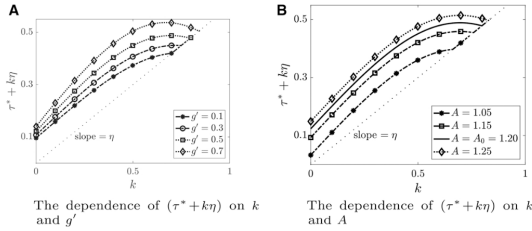


Figure 3
Bankruptcy threshold and the firm's lifespan: g' and A
We plot the time to bankruptcy $\tau^* + k\eta$ against the bankruptcy threshold k , and vary the post-shock growth rate g' in panel A and the asset-to-debt ratio A in panel B. In both panels, the kinks mark the thresholds beyond which creditors exit immediately ($\tau^* = 0$) and the curves become linear ($k\eta$). When either g' or A is sufficiently high, the optimal bankruptcy threshold is interior. The parameters are $A=1.2$, $g=2$, $\lambda=0.2$, $\eta=0.6$ (panel A), and $g=2$, $\lambda=0.2$, $\eta=0.6$, $g'=0.5$ (panel B).

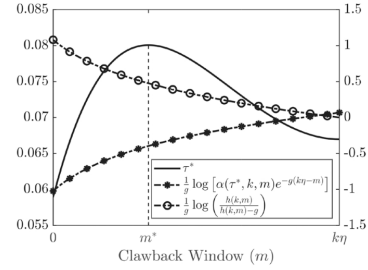


Figure 4
Decomposition of the equilibrium waiting time τ^* .
We decompose the equilibrium waiting time (τ^*) into the recovery rate channel (i.e., $\frac{1}{g} \log(\alpha e^{-g(k\eta-m)})$ with dots) and the hazard rate channel (i.e., $\frac{1}{g} \log(\frac{h(k,m)}{h(k,m)-g})$ with circles) according to the two terms in (30). The curves of τ^* correspond to the left y-axis, and the remaining two curves correspond to the right y-axis. The parameters for this figure are $A=1.1$, $g=2$, $\lambda=0.5$, $\eta=1$, $g'=0.1$, and $k=0.5$.

5.4 Figure 4: optimal clawback window

Read:

- Figure 4 plots equilibrium waiting time against the clawback window m .
- The recovery-rate channel increases in m because more repayments are seized and redistributed.
- The hazard-rate channel decreases in m because it becomes harder to exit successfully before entering the clawback window.
- Their combination is hump-shaped, giving an interior optimal clawback window.

Economic message. Too little clawback does not preserve enough bankruptcy value; too much clawback makes creditors panic because successful exit becomes too difficult.

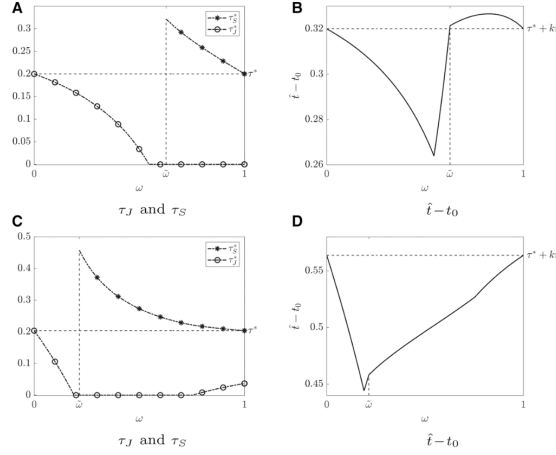


Figure 5
Equilibrium waiting time and time to bankruptcy with heterogeneous creditors
This figure depicts the equilibrium waiting times (τ_J^* , τ_S^*) in panels A and C and the firm's lifespan $\hat{t} - t_0$ (defined in (36)) in panels B and D depending on the fraction of the senior creditors ω . Senior creditors are more patient than junior ones. An initial increase in ω shortens the lifespan because less patient junior creditors trigger bankruptcy. As ω increases further, senior creditors start to exit before bankruptcy, and their patience prolongs the firm's survival relative to the case of a single creditor class. These plots demonstrate the nonmonotonic effects of creditor heterogeneity on coordination and time to bankruptcy. The parameters are $A=1.1$, $g=2$, $\lambda=0.2$, $\eta=0.6$, $g'=1$, $k=0.2$ (panels A and B), and $k=0.6$ (panels C and D).

5.5 Figure 5: seniority and nonmonotonic coordination

Read:

- Figure 5 shows waiting times for junior and senior creditors and the resulting time to bankruptcy as the senior share ω changes.
- Senior creditors are always more patient than juniors.
- A small rise in the senior share shortens firm life because junior creditors get worse recovery and run earlier.
- At higher ω , seniors become pivotal and their patience can increase firm life.

Economic message. Seniority is not automatically stabilizing. Its effect depends on who is pivotal in the run.

5.6 What is missing from the figures

- This is mainly a theory paper, so the visual evidence consists of numerical illustrations rather than empirical tables.
- The figures are therefore not “reduced-form evidence” in the usual sense. Their job is to clarify the logic of Propositions 1–10 and to show where the theoretical forces dominate each other.

6 Short summary

- **Method.** The paper builds a dynamic creditor-run model with asynchronous information, endogenous exit timing, a bankruptcy threshold chosen by the firm, and a clawback window chosen by a regulator.

- **Core mechanism.** Creditors decide how long to stay invested by trading off interest income against the risk of being caught in bankruptcy. Their waiting time is determined by a recovery-rate channel and a hazard-rate channel.
- **Main result 1.** Committing to earlier bankruptcy filing can delay bankruptcy because it preserves assets for latecomers and improves bankruptcy recovery.
- **Main result 2.** Avoidable preference / repayment clawback can outperform early filing because it shifts forward the effective pivotal creditor without shutting down production as early.
- **Main result 3.** The welfare-maximizing regulator wants earlier intervention than the firm, and the optimal clawback window is a relatively detail-free function of contractual and information parameters.
- **Main extension.** Seniority has a nonmonotonic effect on coordination: a small number of senior creditors worsens runs, but a sufficiently large senior class can delay bankruptcy.
- **Policy lesson.** Bankruptcy rules must be judged by how they affect *ex ante* run incentives, not just by how they allocate value *ex post*.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that bankruptcy regulations affect creditor coordination through both recovery and hazard.
- It shows that preserving more value in bankruptcy can make creditors more willing to stay invested before bankruptcy occurs.
- It also shows that *ex post* clawback can be strictly better than the firm’s own commitment to early bankruptcy filing, because clawback preserves production longer while delivering similar discipline on exits.

7.2 Economic intuition

- A creditor waits longer when bankruptcy is less painful.
- But a creditor waits less when bankruptcy is more likely to arrive before exit.
- Regulations such as automatic stay and avoidable preference therefore work only through the balance between those two forces.
- The subtle point is that “more protection in bankruptcy” can either improve or worsen coordination depending on whether the gain in recovery outweighs the extra hazard.

7.3 Takeaway

This paper is important because it changes how one should think about bankruptcy regulation. The right question is not simply whether a rule coordinates creditors in court. The right question is whether the rule makes creditors more willing to keep funding a distressed but still productive firm

before the court ever becomes involved. In the model, both automatic stay and avoidable preference can do that, but avoidable preference is the more powerful instrument because it preserves value while still reshaping the run incentives of creditors.